

PORT ARTHUR, TX, USA

WMO: 722410

Lat: 29.951N

Lon: 94.021W

Elev: 16

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12917

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.4	34.9	17.2	13.1	39.3	22.2	16.7	43.6	24.6	57.2	22.4	56.1	7.8	20	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.5	95.1	77.8	93.4	78.0	91.9	78.0	81.5	89.2	80.5	88.1	80.0	87.5	7.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	152.8	84.8	78.7	148.8	84.0	78.0	145.5	83.3	45.0	88.8	44.1	88.6	43.3	87.6	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.8	19.3	17.7	26.5	98.6	3.2	2.8	24.2	100.6	22.3	102.2	20.5	103.8	18.1	105.8		
			23.5	83.1	3.1	1.9	21.2	84.5	19.4	85.6	17.7	86.7	15.4	88.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.8	53.5	57.3	63.0	69.2	76.5	81.8	83.2	83.5	79.7	71.4	62.1	55.3		
	DBStd	12.71	9.54	9.25	7.95	6.20	4.47	2.87	2.27	2.74	4.32	7.07	8.41	9.37		
	HDD50	176	73	35	9	0	0	0	0	0	0	0	9	50		
	HDD65	1286	372	243	133	32	2	0	0	0	0	27	152	325		
	CDD50	7388	181	239	411	575	820	953	1030	1037	892	662	372	216		
	CDD65	3024	15	27	71	157	357	503	565	572	442	224	65	26		
	CDH74	28307	25	60	241	871	2970	5322	6240	6279	4181	1732	316	70		
	CDH80	11036	0	2	15	118	892	2262	2744	2834	1661	475	32	1		
(14) Wind	WSAvg	8.3	9.5	10.0	10.0	10.1	9.2	7.6	5.9	5.8	6.8	7.8	8.4	9.1		
(15) Precipitation	PrecAvg	58.1	5.2	3.5	3.3	3.7	5.2	6.0	5.1	5.8	6.3	4.6	4.8	4.8		
	PrecMax	81.6	14.9	7.7	9.4	15.3	13.2	18.9	15.7	17.2	22.0	15.1	10.8	18.0		
	PrecMin	34.4	0.6	0.2	0.2	0.3	0.0	0.7	1.5	0.9	1.0	0.0	0.2	1.9		
	PrecStd	12.0	3.1	2.0	2.1	2.9	3.7	4.3	2.8	4.3	4.5	3.9	2.7	3.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	78.9	82.0	85.7	92.0	95.8	96.2	98.4	96.1	90.2	83.3	79.1		
		MCWB	66.6	67.7	68.9	71.2	74.2	76.5	77.9	77.6	76.6	75.8	73.2	70.7		
	2%	DB	74.0	75.8	79.1	83.0	88.4	93.2	94.2	95.4	92.9	87.6	80.4	76.0		
		MCWB	66.0	67.6	68.4	70.8	74.6	77.1	78.4	78.4	76.7	75.1	71.3	69.7		
	5%	DB	71.2	73.1	76.8	81.1	86.6	91.4	92.6	93.4	90.6	85.0	77.7	73.0		
		MCWB	65.1	66.2	67.7	70.2	74.7	77.4	78.6	78.5	76.5	73.4	69.2	68.1		
	10%	DB	68.2	70.4	74.3	79.0	85.0	89.8	91.1	91.6	88.5	82.5	75.1	70.2		
		MCWB	63.4	65.4	66.8	69.6	74.4	77.3	78.5	78.6	75.9	72.4	68.3	66.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.5	71.4	72.9	76.6	80.5	82.4	82.4	82.6	81.1	79.8	75.7	73.3		
		MCDB	73.5	75.9	77.6	81.1	85.2	88.9	90.9	90.4	86.9	84.7	80.0	76.4		
	2%	WB	69.0	70.1	71.4	74.9	78.4	80.5	81.2	81.5	79.9	78.2	73.8	71.3		
		MCDB	71.8	73.6	75.5	79.2	83.7	87.5	89.6	89.5	86.7	83.2	77.7	74.5		
	5%	WB	67.2	68.6	70.3	73.5	77.2	79.7	80.3	80.5	79.0	76.8	72.2	69.6		
		MCDB	69.9	71.4	74.0	77.9	83.2	86.8	88.5	88.4	86.0	81.7	75.9	72.3		
	10%	WB	64.8	66.9	69.1	72.4	76.1	78.8	79.7	79.9	78.1	75.4	70.3	67.1		
		MCDB	67.2	69.5	72.5	76.5	82.1	85.9	87.6	87.8	84.7	80.1	74.0	69.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	17.4	17.8	17.6	16.4	15.9	16.0	16.5	16.9	18.8	19.0	17.9		
		MCDBR	19.3	17.7	17.2	17.8	17.3	18.1	18.2	19.1	18.8	18.7	18.8	18.3		
		MCWBR	12.9	11.0	9.2	8.6	6.9	6.0	6.1	5.9	6.4	8.3	10.9	12.7		
	5% WB	MCDBR	15.5	14.9	14.9	13.6	14.0	15.1	16.4	16.7	15.1	14.9	16.7	16.2		
		MCWBR	12.2	10.9	9.7	8.2	6.8	6.4	6.4	6.4	6.2	8.4	12.0	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.342	0.358	0.376	0.433	0.478	0.491	0.501	0.482	0.457	0.385	0.372	0.351		
	taud		2.518	2.482	2.424	2.263	2.170	2.179	2.186	2.242	2.278	2.464	2.441	2.496		
	Ebn at Noon		283	288	289	274	261	256	253	257	260	275	269	273		
	Edh at Noon		28	32	36	43	48	47	47	44	41	32	30	28		
(44) All-Sky Solar Radiation	RadAvg		913	1032	1435	1727	1938	1984	1899	1800	1584	1390	1051	807		
	RadStd		88	172	127	123	131	147	158	152	160	176	104	91		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.67	N/A	-0.63	N/A	N/A	N/A	+173
(47) Regional (33 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	+162

Nomenclature: See separate page